

# Multi-dimensional Runge Kutta Stability

CPS 406

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# Introduction

- **Context:** Many engineering problems are modeled using ordinary differential equations. When a problem has more than one changing quantity, we write it in the form of system of differential equations.
- **Problem Statement:** If the exact solution of a system should decay smoothly, a numerical method can become unstable if the step size is too large.
- **Objective:** This presentation shows how step size affects stability for a simple two-dimensional Ordinary Differential Equation system using Euler's method and the fourth-order Runge-Kutta method.

## Core Concept

Stability asks whether the numerical solution behaves like the exact solution. If the exact solution decays, the numerical solution should also decay instead of growing.

# Two dimensional ODE System

For  $y_1' = -2y_1$ , (initial :  $y_1(0) = 1$ ):

$$\frac{dy_1}{dt} = -2y_1$$

$$\frac{1}{y_1} dy_1 = -2dt$$

$$\int \frac{1}{y_1} dy_1 = \int -2dt$$

$$\ln |y_1| = -2t + C$$

$$y_1(t) = Ce^{-2t}$$

$$y_1(0) = 1 \implies 1 = Ce^0 \implies C = 1$$

$$\mathbf{y_1(t) = e^{-2t}}$$

For  $y_2' = -5y_2$ , (initial :  $y_2(0) = 1$ ):

$$\frac{dy_2}{dt} = -5y_2$$

$$\frac{1}{y_2} dy_2 = -5dt$$

$$\int \frac{1}{y_2} dy_2 = \int -5dt$$

$$\ln |y_2| = -5t + C$$

$$y_2(t) = Ce^{-5t}$$

$$y_2(0) = 1 \implies 1 = Ce^0 \implies C = 1$$

$$\mathbf{y_2(t) = e^{-5t}}$$

## Expected Behavior

Since both exact solutions decay to zero ( $e^{-2t}, e^{-5t} \rightarrow 0$ ), any stable numerical method must also produce decaying values.

# ODE Methods as Numerical Integration

To show you that: ODE solving  $\approx$  approximating an integral

① **Start with the ODE:**  $y'(t) = f(t, y)$

② **Integrate in the time step**  $[t_n, t_{n+1}]$ : Let  $h(\text{step size}) = t_{n+1} - t_n$ . Then:

$$\int_{t_n}^{t_{n+1}} y'(t) dt = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

③ **Apply the Fundamental Theorem of Calculus:** The left-hand side simplifies directly:

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

## The Fundamental Integral Equation

After adding  $y(t_n)$  on both sides, we get the basis for all one-step numerical methods:

$$y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(t, y(t)) dt$$

# Euler's Method for the 2D System

**Intuition:** Euler's method approximates the integral by assuming that the slope  $f(t, y)$  forms a constant rectangle of height  $f(t_n, y_n)$  over width  $h$ . Since area of the rectangle  $\Rightarrow w \times h \Rightarrow hf(t_n, y_n)$

## Step 1: Vector Form

Define the discrete state and slope vectors from our original ODE system:

$$\mathbf{y}_n = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix}, \quad \mathbf{f}(t_n, \mathbf{y}_n) = \begin{bmatrix} -2y_{1,n} \\ -5y_{2,n} \end{bmatrix}$$

## Step 2: Substitution

Substituting these vectors into our formula  $\mathbf{y}_{n+1} = \mathbf{y}_n + hf(t_n, \mathbf{y}_n)$ , we get:

$$\begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix} + h \begin{bmatrix} -2y_{1,n} \\ -5y_{2,n} \end{bmatrix}$$

## Step 3: Component-wise Addition

Distributing step size  $h$  and then adding component by component into one system:

$$\begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} - 2hy_{1,n} \\ y_{2,n} - 5hy_{2,n} \end{bmatrix}$$

## Step 4: Factor

Further factoring will yield two formulas. The multipliers will be later important for stability:

## Final Update Formulas

$$y_{1,n+1} = (1 - 2h)y_{1,n}$$

$$y_{2,n+1} = (1 - 5h)y_{2,n}$$

# Euler Stability Analysis: Component 1

## The Stability Idea:

From our Euler formula update, we found that each step multiplies the old value by a specific factor:

$$y_{1,n+1} = (1 - 2h)y_{1,n}, \quad y_{2,n+1} = (1 - 5h)y_{2,n}$$

For the exact solutions ( $e^{-2t}$  and  $e^{-5t}$ ), we saw that both components decay to zero. So, for the numerical method to decay, the multipliers must have an absolute value of less than 1:

$$|1 - 2h| < 1 \quad \text{and} \quad |1 - 5h| < 1$$

## First Condition ( $y_1$ )

$$|1 - 2h| < 1$$

$$-1 < 1 - 2h < 1 \quad (\text{Expanding})$$

$$-2 < -2h < 0 \quad (\text{Subtracting 1 on both sides})$$

$$1 > h > 0 \quad (\text{Dividing by } -2, \text{ and flipping inequality signs})$$

Rewriting in increasing order, the first component is stable when:

$$0 < h < 1$$

# Euler Stability Analysis: Component 2

## Second Condition ( $y_2$ )

$$|1 - 5h| < 1$$

$$-1 < 1 - 5h < 1 \quad (\text{Expanding})$$

$$-2 < -5h < 0 \quad (\text{Subtracting 1 on both sides})$$

$$\frac{2}{5} > h > 0 \quad (\text{Dividing by } -5, \text{ and flipping inequality signs})$$

Rewriting in increasing order, the second component is stable when:

$$0 < h < 0.4$$

## Combine Both Conditions:

The full system is stable only if *both* components are stable simultaneously.

- First component requires:  $0 < h < 1$  and Second component requires:  $0 < h < 0.4$

The stricter condition dominates. So, Euler's method is stable for the whole 2D system only when:

$$0 < h < 0.4$$

# Euler Example: Stable Step Size

So, the stability condition:  $0 < h < 0.4$ . Let's test a step size,  $h = 0.1$ .

## 1. Calculate Multipliers

Substituting  $h = 0.1$  into our update formulas:

$$\begin{aligned}y_{1,n+1} &= (1 - 2(0.1))y_{1,n} \\ &= (1 - 0.2)y_{1,n}\end{aligned}$$

$$\mathbf{y}_{1,n+1} = \mathbf{0.8y}_{1,n}$$

$$\begin{aligned}y_{2,n+1} &= (1 - 5(0.1))y_{2,n} \\ &= (1 - 0.5)y_{2,n}\end{aligned}$$

$$\mathbf{y}_{2,n+1} = \mathbf{0.5y}_{2,n}$$

## 2. Iterate the System

Start with our initial condition  $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ :

$$\mathbf{y}_1 = \begin{bmatrix} 0.8(1) \\ 0.5(1) \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}$$

$$\mathbf{y}_2 = \begin{bmatrix} 0.8(0.8) \\ 0.5(0.5) \end{bmatrix} = \begin{bmatrix} 0.64 \\ 0.25 \end{bmatrix}$$

$$\mathbf{y}_3 = \begin{bmatrix} 0.8(0.64) \\ 0.5(0.25) \end{bmatrix} = \begin{bmatrix} 0.512 \\ 0.125 \end{bmatrix}$$

## Result

Both components strictly decay toward zero. The numerical method is **stable**.

# Euler Example: Unstable Step Size

Now let's test  $h = 0.5$ .

## 1. Calculate Multipliers

Substitute  $h = 0.5$  into our update formulas:

$$\begin{aligned}y_{1,n+1} &= (1 - 2(0.5))y_{1,n} \\ &= (1 - 1)y_{1,n}\end{aligned}$$

$$\mathbf{y}_{1,n+1} = \mathbf{0} \cdot \mathbf{y}_{1,n} = \mathbf{0}$$

$$\begin{aligned}y_{2,n+1} &= (1 - 5(0.5))y_{2,n} \\ &= (1 - 2.5)y_{2,n}\end{aligned}$$

$$\mathbf{y}_{2,n+1} = -1.5\mathbf{y}_{2,n}$$

## 2. Iterate the System

Start with initial condition  $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ :

$$\mathbf{y}_1 = \begin{bmatrix} 0(1) \\ -1.5(1) \end{bmatrix} = \begin{bmatrix} 0 \\ -1.5 \end{bmatrix}$$

$$\mathbf{y}_2 = \begin{bmatrix} 0(0) \\ -1.5(-1.5) \end{bmatrix} = \begin{bmatrix} 0 \\ 2.25 \end{bmatrix}$$

$$\mathbf{y}_3 = \begin{bmatrix} 0(0) \\ -1.5(2.25) \end{bmatrix} = \begin{bmatrix} 0 \\ -3.375 \end{bmatrix}$$

## Result

The second component's magnitude grows ( $1 \rightarrow -1.5 \rightarrow 2.25 \rightarrow -3.375 \dots$ ). The numerical method is **unstable**.

## RK4 Derivation: Computing $\mathbf{k}_1$ and $\mathbf{k}_2$

**System Setup:**  $\mathbf{f}(t, \mathbf{y}) = \begin{bmatrix} -2y_1 \\ -5y_2 \end{bmatrix}$ , Initial:  $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ , Step size:  $h = 0.1$

**Step 1: Computing  $\mathbf{k}_1$**

$$\mathbf{k}_1 = \mathbf{f}(t_0, \mathbf{y}_0) = \begin{bmatrix} -2(1) \\ -5(1) \end{bmatrix} = \boxed{\begin{bmatrix} -2 \\ -5 \end{bmatrix}}$$

**Step 2: Computing  $\mathbf{k}_2$**

Formula:  $\mathbf{k}_2 = \mathbf{f}\left(t_0 + \frac{h}{2}, \mathbf{y}_0 + \frac{h}{2}\mathbf{k}_1\right)$

$$\frac{h}{2}\mathbf{k}_1 = 0.05 \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} -0.1 \\ -0.25 \end{bmatrix}$$

$$\mathbf{y}_0 + \frac{h}{2}\mathbf{k}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.1 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0.75 \end{bmatrix}$$

$$\mathbf{k}_2 = \mathbf{f}\left(t_0 + 0.05, \begin{bmatrix} 0.9 \\ 0.75 \end{bmatrix}\right) = \begin{bmatrix} -2(0.9) \\ -5(0.75) \end{bmatrix} = \boxed{\begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix}}$$

# RK4 Derivation: Computing $\mathbf{k}_3$ and $\mathbf{k}_4$

## Step 3: Computing $\mathbf{k}_3$

Formula:  $\mathbf{k}_3 = \mathbf{f}\left(t_0 + \frac{h}{2}, \mathbf{y}_0 + \frac{h}{2}\mathbf{k}_2\right)$

$$\frac{h}{2}\mathbf{k}_2 = 0.05 \begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix} = \begin{bmatrix} -0.09 \\ -0.1875 \end{bmatrix}$$

$$\mathbf{y}_0 + \frac{h}{2}\mathbf{k}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.09 \\ -0.1875 \end{bmatrix} = \begin{bmatrix} 0.91 \\ 0.8125 \end{bmatrix}$$

## Step 4: Computing $\mathbf{k}_4$

Formula:  $\mathbf{k}_4 = \mathbf{f}(t_0 + h, \mathbf{y}_0 + h\mathbf{k}_3)$

$$\mathbf{k}_3 = \begin{bmatrix} -2(0.91) \\ -5(0.8125) \end{bmatrix} = \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix}$$

$$h\mathbf{k}_3 = 0.1 \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix} = \begin{bmatrix} -0.182 \\ -0.40625 \end{bmatrix}$$

$$\mathbf{y}_0 + h\mathbf{k}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.182 \\ -0.40625 \end{bmatrix} = \begin{bmatrix} 0.818 \\ 0.59375 \end{bmatrix}$$

$$\mathbf{k}_4 = \begin{bmatrix} -2(0.818) \\ -5(0.59375) \end{bmatrix} = \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix}$$

# RK4 Derivation: Combining Slopes

## Step 5: Computing final weighted sum for $y_1$

Formula:  $y_1 = y_0 + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$\begin{aligned}\sum \mathbf{k} &= \begin{bmatrix} -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} -1.8 \\ -3.75 \end{bmatrix} + 2 \begin{bmatrix} -1.82 \\ -4.0625 \end{bmatrix} + \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ -5 \end{bmatrix} + \begin{bmatrix} -3.6 \\ -7.5 \end{bmatrix} + \begin{bmatrix} -3.64 \\ -8.125 \end{bmatrix} + \begin{bmatrix} -1.636 \\ -2.96875 \end{bmatrix} = \begin{bmatrix} -10.876 \\ -23.59375 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\mathbf{y}_1 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{0.1}{6} \begin{bmatrix} -10.876 \\ -23.59375 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -0.18127 \\ -0.39323 \end{bmatrix} = \boxed{\begin{bmatrix} 0.81873 \\ 0.60677 \end{bmatrix}}\end{aligned}$$

## Accuracy Check at $t = 0.1$

Exact Solutions:  $y_1(0.1) = e^{-0.2} \approx 0.81873$ ,  $y_2(0.1) = e^{-0.5} \approx 0.60653$

RK4 Estimate:  $y_1 \approx 0.81873$ ,  $y_2 \approx 0.60677$

*RK4 is highly accurate after just one step!*

# RK4 Stability

To study stability, we use the standard linear test equation:

$$y' = \lambda y \implies f(t, y) = \lambda y$$

The 4th-order Runge-Kutta formulas are:

$$k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

## RK4 Stability: Computing $k_1$ and $k_2$

Substitute  $f(t, y) = \lambda y$  into the RK4 slopes:

### Step 1: Computing $k_1$

$$k_1 = \lambda y_n$$

### Step 2: Computing $k_2$

We substitute  $k_1$ :

$$\begin{aligned} k_2 &= \lambda \left( y_n + \frac{h}{2} k_1 \right) \\ &= \lambda \left( y_n + \frac{h}{2} \lambda y_n \right) \\ &= \lambda y_n \left( 1 + \frac{h\lambda}{2} \right) \\ \mathbf{k}_2 &= \lambda \mathbf{y}_n \left( \mathbf{1} + \frac{\mathbf{z}}{2} \right) \end{aligned}$$

Define  $z = h\lambda$ :

# RK4 Stability: Computing $k_3$

## Step 3: Computing $k_3$

Substituting  $k_2$  and use  $\frac{h\lambda}{2} = \frac{z}{2}$ :

$$\begin{aligned}k_3 &= \lambda \left( y_n + \frac{h}{2} k_2 \right) \\&= \lambda \left[ y_n + \frac{h}{2} \lambda y_n \left( 1 + \frac{z}{2} \right) \right] \\&= \lambda y_n \left[ 1 + \frac{z}{2} \left( 1 + \frac{z}{2} \right) \right]\end{aligned}$$

Expanding:

$$\mathbf{k}_3 = \lambda \mathbf{y}_n \left( \mathbf{1} + \frac{\mathbf{z}}{2} + \frac{\mathbf{z}^2}{4} \right)$$

## RK4 Stability: Computing $k_4$

### Step 4: Computing $k_4$

Substituting  $k_3$  and use  $h\lambda = z$ :

$$\begin{aligned}k_4 &= \lambda(y_n + hk_3) \\&= \lambda \left[ y_n + h\lambda y_n \left( 1 + \frac{z}{2} + \frac{z^2}{4} \right) \right] \\&= \lambda y_n \left[ 1 + z \left( 1 + \frac{z}{2} + \frac{z^2}{4} \right) \right]\end{aligned}$$

Expanding:

$$\mathbf{k}_4 = \lambda \mathbf{y}_n \left( \mathbf{1} + \mathbf{z} + \frac{\mathbf{z}^2}{2} + \frac{\mathbf{z}^3}{4} \right)$$

# RK4 Stability: Final Substitution

## Step 5: Substituting into RK4 update

Recall  $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ . Substitute the  $k$ 's:

$$\begin{aligned}y_{n+1} &= y_n + \frac{h}{6}\lambda y_n \left[ 1 + 2\left(1 + \frac{z}{2}\right) + 2\left(1 + \frac{z}{2} + \frac{z^2}{4}\right) + \left(1 + z + \frac{z^2}{2} + \frac{z^3}{4}\right) \right] \\&= y_n + \frac{z}{6}y_n \left[ 1 + (2 + z) + \left(2 + z + \frac{z^2}{2}\right) + \left(1 + z + \frac{z^2}{2} + \frac{z^3}{4}\right) \right]\end{aligned}$$

Combine like terms:

- **Constants:**  $1 + 2 + 2 + 1 = 6$
- **$z$ -terms:**  $z + z + z = 3z$
- **$z^2$ -terms:**  $\frac{z^2}{2} + \frac{z^2}{2} = z^2$
- **$z^3$ -term:**  $\frac{z^3}{4}$

$$y_{n+1} = y_n + y_n \frac{z}{6} \left( 6 + 3z + z^2 + \frac{z^3}{4} \right)$$

# RK4 Stability: The Multiplier

## Step 6: Further distributing $\frac{z}{6}$

$$\begin{aligned}y_{n+1} &= y_n + y_n \left[ \frac{z}{6}(6) + \frac{z}{6}(3z) + \frac{z}{6}(z^2) + \frac{z}{6} \left( \frac{z^3}{4} \right) \right] \\&= y_n + y_n \left( z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \right)\end{aligned}$$

Factor out  $y_n$ :

$$y_{n+1} = y_n \left( 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \right)$$

So,  $y_{n+1} = R(z)y_n$ , where the RK4 stability multiplier is:

## RK4 Stability Function

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

# RK4 Stability Condition

For the numerical solution to decay and remain stable, the multiplier must have an absolute value strictly less than 1:

$$|R(z)| < 1$$

For our two-dimensional system:

$$y_1' = -2y_1 \implies \lambda_1 = -2$$

$$y_2' = -5y_2 \implies \lambda_2 = -5$$

Since  $z = h\lambda$ , we have:

$$z_1 = -2h$$

$$z_2 = -5h$$

So, RK4 is stable for the system when **both** conditions are true:

$$|R(-2h)| < 1$$

and

$$|R(-5h)| < 1$$

## RK4 Example with $h = 0.5$

Earlier, we saw that the Euler method was unstable at  $h = 0.5$ . Let's check if RK4 is stable or not:

RK4 stability condition:

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} \quad \text{where} \quad |R(z)| < 1$$

For our 2D system ( $y_1' = -2y_1$ ,  $y_2' = -5y_2$ ), we have:

$$\lambda_1 = -2, \quad \lambda_2 = -5$$

With  $h = 0.5$  and  $z = h\lambda$ , we evaluate:

$$z_1 = 0.5(-2) = -1$$

$$z_2 = 0.5(-5) = -2.5$$

We must check both  $|R(-1)| < 1$  and  $|R(-2.5)| < 1$ .

## RK4 Example with $h = 0.5$ : Compute $R(-1)$

Substituting  $z = -1$  into the stability multiplier:

$$\begin{aligned} R(-1) &= 1 + (-1) + \frac{(-1)^2}{2} + \frac{(-1)^3}{6} + \frac{(-1)^4}{24} \\ &= 1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \\ &= \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \end{aligned}$$

LCM:

$$\begin{aligned} R(-1) &= \frac{12}{24} - \frac{4}{24} + \frac{1}{24} \\ &= \frac{9}{24} = \frac{3}{8} \end{aligned}$$

### Component 1 Stability

$$\boxed{R(-1) = 0.375} \implies |0.375| < 1$$

The first component is **stable**.

## RK4 Example with $h = 0.5$ : Compute $R(-2.5)$

Substituting  $z = -2.5$  into the stability multiplier:

$$R(-2.5) = 1 + (-2.5) + \frac{(-2.5)^2}{2} + \frac{(-2.5)^3}{6} + \frac{(-2.5)^4}{24}$$

Calculation:

- $(-2.5)^2 = 6.25 \implies \frac{6.25}{2} = 3.125$
- $(-2.5)^3 = -15.625 \implies \frac{-15.625}{6} \approx -2.6042$
- $(-2.5)^4 = 39.0625 \implies \frac{39.0625}{24} \approx 1.6276$

Substituting back into the equation:

$$R(-2.5) \approx 1 - 2.5 + 3.125 - 2.6042 + 1.6276$$

## RK4 Example with $h = 0.5$ : Final Conclusion

Combine the terms step by step for  $R(-2.5)$ :

$$1 - 2.5 = -1.5$$

$$-1.5 + 3.125 = 1.625$$

$$1.625 - 2.6042 = -0.9792$$

$$-0.9792 + 1.6276 = 0.6484$$

### Component 2 Stability

$$\boxed{R(-2.5) \approx 0.6484} \implies |0.6484| < 1$$

The second component is also **stable**.

### Comparison with Euler

Both components are stable under RK4 at this step size.

$$\boxed{\text{Euler is unstable at } h = 0.5} \quad \text{vs.} \quad \boxed{\text{RK4 is stable at } h = 0.5}$$

# Extension to a 3D System

Let us consider a 3-dimensional system:

$$\begin{cases} y_1' = -2y_1 \\ y_2' = -5y_2 \\ y_3' = -8y_3 \end{cases}$$

For the **Euler stability**:

$$|1 - 2h| < 1 \implies 0 < h < 1$$

$$|1 - 5h| < 1 \implies 0 < h < 0.4$$

$$|1 - 8h| < 1 \implies 0 < h < 0.25$$

For the entire 3D system to be stable, the step size  $h$  must satisfy the strictest condition:

$$0 < h < 0.25$$

## Main Point

Adding a faster-decaying component can force a smaller stable step size.

- **Summary:** Euler's method is simple but it can become unstable when the step size is too big, and so fourth-order Runge Kutta uses four slope estimates and can remain stable for a larger step size. So, In a **multidimensional system**, every component must satisfy the stability condition.

**Thank you!**